

The Complex Integral

Complex Analysis

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Definition

If $I := [a, b]$ is a closed, bounded interval in $\mathbb{R}$, then a **partition** of I is a finite, ordered set $\mathcal{P} := (x_0, x_1, \dots, x_{n-1}, x_n)$ of points in I such that

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b.$$

The points of $\mathcal{P}$ are used to divide $I = [a, b]$ into non-overlapping subintervals

$$I_1 := [x_0, x_1], \quad I_2 := [x_1, x_2], \dots, \quad I_n := [x_{n-1}, x_n].$$

Often we will denote the partition $\mathcal{P}$ by the notation $\mathcal{P} = \{[x_{i-1}, x_i]\}_{i=1}^n$. We define the **norm** (or **mesh**) of $\mathcal{P}$ to be the number

$$\|\mathcal{P}\| := \sup\{x_1 - x_0, x_2 - x_1, \dots, x_n - x_{n-1}\}.$$

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Definition

If a point x_i^* has been selected from each subinterval $I_i = [x_{i-1}, x_i]$, for $i = 1, 2, \dots, n$, then the points are called **tags** of the subintervals I_i . A set of ordered pairs

$$\mathcal{P}^* := \{(I_i, x_i^*)\}_{i=1}^n$$

of subintervals and corresponding tags is called a **tagged partition** of I .



Figure 1: A tagged partition of $[a, b]$

Definition

If $\mathcal{P}^*$ is a tagged partition of I , we define the **Riemann sum** of a function $f : [a, b] \rightarrow \mathbb{C}$ corresponding to $\mathcal{P}^*$ to be the number

$$S(f; \mathcal{P}^*) := \sum_{i=1}^n f(x_i^*)(x_i - x_{i-1}).$$

Definition

A function $f : [a, b] \rightarrow \mathbb{C}$ is said to be **Riemann integrable** on $[a, b]$ if there exists a number $L \in \mathbb{C}$ such that for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that if $\mathcal{P}^*$ is any tagged partition of $[a, b]$ with $\|\mathcal{P}^*\| < \delta_\varepsilon$, then

$$|S(f; \mathcal{P}^*) - L| < \varepsilon.$$

The set of all Riemann integrable functions on $[a, b]$ is denoted by $\mathcal{R}[a, b]$.

Remark

It is sometimes said that the integral L is the “limit” of the Riemann sums $S(f; \mathcal{P}^)$ as the norm $\|\mathcal{P}^*\| \rightarrow 0$, in symbols*

$$L = \lim_{\|\mathcal{P}^*\| \rightarrow 0} S(f; \mathcal{P}^*) = \lim_{\|\mathcal{P}^*\| \rightarrow 0} \sum_{i=1}^n f(x_i^*)(x_i - x_{i-1}).$$

However, since $S(f; \mathcal{P}^)$ is not a function of $\|\mathcal{P}^*\|$, this **limit** is not of the type we have studied before.*

Proposition

If $f \in \mathcal{R}[a, b]$ with integral L and if $(\mathcal{P}_n^\star)$ is a sequence of tagged partitions of $[a, b]$ such that $\|\mathcal{P}_n^\star\| \rightarrow 0$, then

$$L = \lim S(f; \mathcal{P}_n^\star)$$

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Theorem

If $f \in \mathcal{R}[a, b]$, then the value of L is uniquely determined.

Definition

If $f \in \mathcal{R}[a, b]$, then the number L is called the **Riemann integral** of f (or simply, the **integral**) over $[a, b]$. Instead of L , we will usually write

$$L = \int_a^b f \quad \text{or} \quad \int_a^b dx f(x).$$

It should be understood that any letter other than x can be used in the latter expression, so long as it does not cause ambiguity.

Example

Let $\alpha, \beta \in \mathbb{C}$ and let $x \in \mathbb{R}$. Show that any linear function

$$f(x) = \alpha x + \beta$$

is in $\mathcal{R}[a, b]$ and hence find its integral on $[a, b]$.

Theorem

If g is Riemann integrable on $[a, b]$ and if $f(x) = g(x)$ except for a finite number of points in $[a, b]$, then f is Riemann integrable and

$$\int_a^b f = \int_a^b g.$$

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Theorem

Suppose that f and g are in $\mathcal{R}[a, b]$. Then if $\lambda \in \mathbb{C}$, the function $\lambda f + g$ is in $\mathcal{R}[a, b]$ and

$$\int_a^b (\lambda f + g) = \lambda \int_a^b f + \int_a^b g.$$

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Theorem

If $f : [a, b] \rightarrow \mathbb{C}$ is continuous on $[a, b]$, then $f \in \mathcal{R}[a, b]$.

Theorem (Additivity Theorem)

Let $f : [a, b] \rightarrow \mathbb{C}$ and let $c \in (a, b)$. Then $f \in \mathcal{R}[a, b]$ if and only if its restrictions to $[a, c]$ and $[c, b]$ are both Riemann integrable. In this case

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Corollary

If $a = c_0 < c_1 < \dots < c_m = b$, then $f \in \mathcal{R}[a, b]$ if and only if the restrictions of f to each of the subintervals $[c_{i-1}, c_i]$ are Riemann integrable. In this case we have

$$\int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f.$$

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Definition

If $f \in \mathcal{R}[a, b]$ and if $\alpha, \beta \in [a, b]$ with $\alpha < \beta$, we define

$$\int_\beta^\alpha f := - \int_\alpha^\beta f \quad \text{and} \quad \int_\alpha^\alpha f := 0.$$

Theorem (Fundamental Theorem of Calculus (First Form))

Suppose that there is a finite set E in $[a, b]$ and functions $f, F := [a, b] \rightarrow \mathbb{C}$ such that:

- a) F is continuous on $[a, b]$,*
- b) $F'(x) = f(x)$ for all $x \in [a, b] \setminus E$,*
- c) f belongs to $\mathcal{R}[a, b]$.*

Then we have

$$\int_a^b f = F(b) - F(a).$$

Theorem (Fundamental Theorem of Calculus (Second Form))

Let $f \in \mathcal{R}[a, b]$ and let f be continuous at a point $c \in [a, b]$. Then the **indefinite integral** F of f with **basepoint** a , defined as

$$F(x) = \int_a^x f \quad x \in [a, b]$$

is differentiable at c and $F'(c) = f(c)$.

Corollary

If f is continuous on $[a, b]$, then the indefinite integral

$$F(x) = \int_a^x f \quad x \in [a, b]$$

is differentiable on $[a, b]$ and $F'(x) = f(x)$ for all $x \in [a, b]$.

Definition

If f is in $\mathcal{R}[a, b]$ for every $b > a$ and $a \in \mathbb{R}$ fixed and if the limit of the indefinite integral F with basepoint a

$$\lim_{b \rightarrow \infty} F(b) = \lim_{b \rightarrow \infty} \int_a^b f$$

exists in $\mathbb{C}$, then the **improper integral**

$$\int_a^\infty dt f(t)$$

is defined to be this limit.

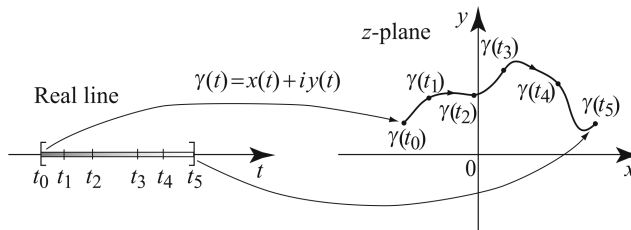
Theorem (Substitution Theorem)

Let $J := [\alpha, \beta]$ and let $\varphi : J \rightarrow \mathbb{R}$ have a continuous derivative on J . If $f : I \rightarrow \mathbb{C}$ is continuous on an interval I containing $\varphi(J)$, then

$$\int_{\alpha}^{\beta} (f \circ \varphi) \varphi' = \int_{\varphi(\alpha)}^{\varphi(\beta)} f.$$

Definition

1. A **curve** is a map $\gamma : I \subseteq \mathbb{R} \rightarrow \mathbb{C}$ from the real interval $I := [a, b]$ into the complex plane; $\gamma(a)$ is called the **initial point** of the curve and $\gamma(b)$ its **final point**, if the former and the latter coincide, then the curve is said to be **closed**. Moreover, γ is said to **join** its initial and final points. The range $\gamma^\dagger = \gamma(I) \subset \mathbb{C}$ is called the **track** of γ , and if $\gamma^\dagger \subset \Omega \subseteq \mathbb{C}$, then we say that γ is a curve in Ω .

Figure 2: Mapping from a real interval I into the complex plane

Definition

2. A **simple curve** is a curve that does not self-intersect, i.e., γ is injective, so that $\gamma(t_1) \neq \gamma(t_2)$ when $t_1 \neq t_2$. When the curve is closed and simple except for the fact that $\gamma(a) = \gamma(b)$, then it is called a **Jordan curve**.
3. A **path** is a finite collection $\{\gamma_i\}_{i=1}^n$ of simple curves such that the initial point of $\gamma_k + 1$ coincides with the final point of γ_k .
4. A **smooth curve** is a continuously differentiable curve for which $\gamma'(t) \neq 0$ for $t \in I$.
5. A **contour** is a path whose curves are smooth. When the initial point of γ_1 coincides with the final point of γ_n , the contour is said to be a **closed contour**.

Remark

By derivative at the endpoint a [respectively, b] we mean the appropriate sided derivative, namely

$$\gamma'(a) = \lim_{z \rightarrow a^+} \frac{\gamma(z) - \gamma(a)}{z - a}, \quad \left[\text{respectively, } \lim_{z \rightarrow b^-} \frac{\gamma(z) - \gamma(b)}{z - b} \right].$$

Remark

Let $\mathcal{P} = \{[t_{k-1}, t_k]\}_{k=1}^n$ be a partition of $I := [a, b]$ and let γ be a smooth curve from I into $\mathbb{C}$. We can approximate the length of γ via sums of the form

$$\sum_{i=1}^n |\gamma(t_k) - \gamma(t_{k-1})|.$$

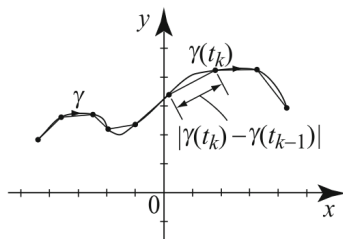


Figure 3: Sum approximation for the length of a curve γ

Definition

Let $\gamma : I \subseteq \mathbb{R} \rightarrow \mathbb{C}$ be a contour and let $\gamma_i : I_i \subset I \rightarrow \mathbb{C}$ be the collection of simple smooth curves as in the previous definition. We define the **length** of γ to be the number

$$L(\gamma) = \sum_{j=1}^n \int_{I_j} |\gamma'|.$$

Example

Find the arc length of a **cycloid** with radius $a > 0$ defined by

$$\gamma(t) = a(t - \sin t) + ia(1 - \cos t), \quad t \in [0, 2\pi].$$

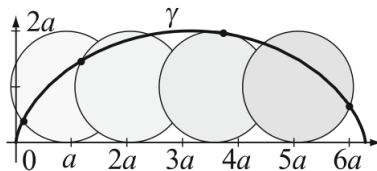


Figure 4: Arc length of a cycloid

Definition

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a contour and let f be continuous on $\gamma^\dagger$. Then the **contour integral of f along γ** is defined to be the number

$$L = \int_a^b (f \circ \gamma) \gamma'.$$

Instead of L , we will usually write

$$L = \int_\gamma f \quad \text{or} \quad \int_\gamma dz f(z).$$

If Γ is a closed contour, the notation

$$\oint_\Gamma f$$

is usually used to express the fact that the integration is along a closed contour.

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Example

Let $w_1, w_2 \in \mathbb{C}$ and let $\gamma : [0, 1] \rightarrow \mathbb{C}$ be the line segment joining w_1 to w_2 . Then compute

$$\int_\gamma dz \, z.$$

Definition

Let $R > 0$ and let $z_0 \in \mathbb{C}$. The particular closed contour $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$ defined by $\gamma(t) = z_0 + Re^{it}$ is called the **positively oriented circle with centre z_0 and radius R** . The notation

$$\oint_{|z-z_0|=R} dz f(z)$$

is sometimes used to denote the integral of f along such contour.

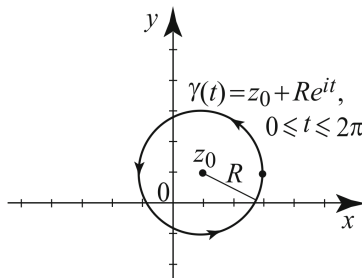


Figure 5: Positively oriented circle with centre z_0 and radius R

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Example

If Γ is the positively oriented circle centred at the origin and with radius $r > 0$, compute the integral

$$\oint_{\Gamma} dz z^n$$

for the different values of $n = -2, -1, 0, 1, 2$.

Theorem (Darboux Inequality)

Suppose f is continuous and bounded on $\gamma^\dagger$, i.e., there exists a positive number M such that $|f(\gamma^\dagger)| \leq M$. Then

$$\left| \int_\gamma f \right| \leq ML(\gamma).$$

More generally

$$\left| \int_\gamma f \right| \leq \int_a^b |f \circ \gamma| |\gamma'|.$$

Definition

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a curve. The curve **opposite** to γ is defined as

$$-\gamma(t) := \gamma(a + b - t)$$

for $t \in [a, b]$.

Theorem

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a contour and let $f, g : \gamma^\dagger \rightarrow \mathbb{C}$ be continuous. Then for all $\lambda \in \mathbb{C}$,

$$\int_\gamma (\lambda f + g) = \lambda \int_\gamma f + \int_\gamma g,$$

and if $-\gamma$ is the contour opposite to γ , then

$$\int_\gamma f = - \int_{-\gamma} f.$$

Definition

Let $\gamma : [a, b] \rightarrow \mathbb{C}$, $\sigma : [\alpha, \beta] \rightarrow \mathbb{C}$ be curves. We say that σ is a **reparametrisation** of γ if there is a continuously differentiable bijection ϕ of $[\alpha, \beta]$ onto $[a, b]$ with an everywhere strictly positive derivative ϕ' , such that $\sigma = \gamma \circ \phi$. Similarly, if $\Gamma : I \subset \mathbb{R} \rightarrow \mathbb{C}$, $\Gamma = \{\Gamma_i\}_{i=1}^n$ and $\Sigma : J \subset \mathbb{R} \rightarrow \mathbb{C}$, $\Sigma = \{\Sigma_i\}_{i=1}^n$ are contours, then we say that Σ is a **reparametrisation** of Γ if Σ_i is a reparametrisation of Γ_i for each $i = 1, \dots, n$.

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Proposition

If Σ is a reparametrisation of Γ , then Σ and Γ have the same track.

Theorem

Let Σ be a reparametrisation of a contour Γ and let $f : \Gamma^\dagger \rightarrow \mathbb{C}$ be continuous.
Then

$$\int_\Sigma f = \int_\Gamma f.$$

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Definition

Let G be an open subset of $\mathbb{C}$ and let $f : G \rightarrow \mathbb{C}$. A function $F \in \mathcal{H}(G)$ is called a **primitive** (or **antiderivative**) for f on G if $F' = f$.

Theorem (The Fundamental Theorem of Contour Integration (First Form))

Suppose that f is continuous on a domain Ω . Then the following statements are equivalent:

- a) f has a primitive on Ω ;
- b) the integral of f along any closed contour in Ω vanishes;
- c) if γ and μ are two contours in Ω joining z_1 to z_2 , then

$$\int_\gamma f = \int_\mu f = F(z_2) - F(z_1),$$

where F is the primitive from a).

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Remark

Due to the fact that if any of the previous statements is true, then the integral along a contour joining z_1 to z_2 is independent of the contour chosen, we will sometimes write

$$\int_{z_1}^{z_2} f$$

to denote the value of such integral.

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Definition

A domain $\Omega \subseteq \mathbb{C}$ is said to be **simply connected** in $\mathbb{C}$ if for any Jordan curve γ in Ω , the interior $\text{int } \gamma$ is also contained in Ω , that is, $\text{int } \gamma \subset \Omega$. We will usually write Ω^1 to express the fact that Ω is a simply connected domain.

Theorem (Green's Theorem)

Let R be a simply connected region whose boundary is the contour γ and let $P, Q : \mathbb{R}^2 \rightarrow \mathbb{R}$ be at least class $C^1[R]$. Then

$$\int_{\gamma} (dx P + dy Q) = \iint_R dx dy \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right),$$

where γ is traversed in the positive sense.

Theorem (Cauchy's Theorem)

If f is holomorphic (with a continuous derivative) in a simply connected domain Ω , and if Γ is a closed contour in Ω , then

$$\oint_{\Gamma} f = 0.$$

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Theorem (Cauchy-Goursat Theorem)

If $f \in \mathcal{H}(\Omega^\circ)$ and if Γ is a closed contour in Ω° , then

$$\oint_{\Gamma} f = 0.$$

Theorem (Cauchy-Goursat Theorem for multiply connected domains)

Let $\{\Omega_k\}_{k=1}^n$ be simply connected domains such that $\Omega_i \cap \Omega_k = \emptyset$ whenever $i \neq j$, whose union is contained in another simply connected region Ω_0 . Let $\Omega = \Omega_0 \setminus \left(\bigcup_{k=1}^n \overline{\Omega_k} \right)$. For $j = 1, \dots, n$, let C_j be disjoint simple closed paths with positive orientation whose interior regions contain $\overline{\Omega_j}$ and assume that C_j are such that $C_i \cap C_j = \emptyset$ whenever $i \neq j$. Let C be a simple closed path with positive orientation in Ω whose interior region contains all C_j . Suppose that $f \in \mathcal{H}(\Omega)$. Then we have

$$\oint_C f = \sum_{i=1}^n \oint_{C_i} f.$$

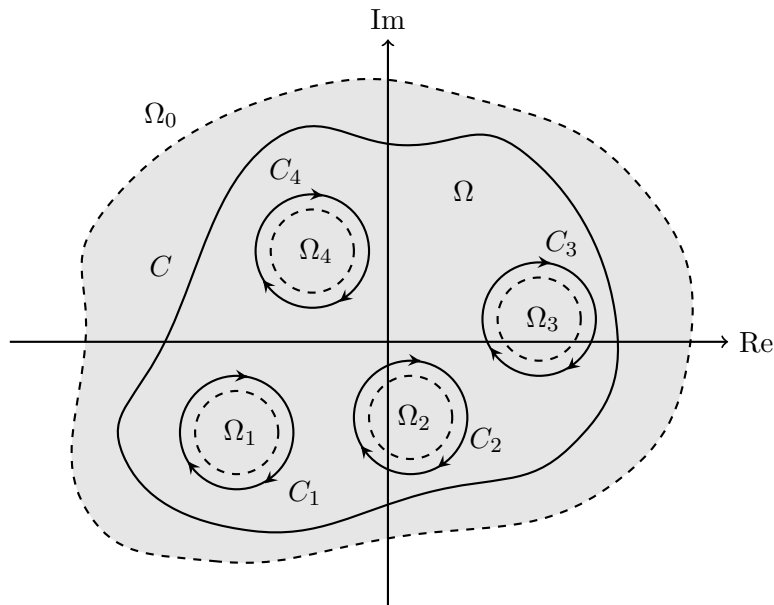


Figure 6: Geometric visualisation of the hypotheses in the Cauchy-Goursat theorem for multiply connected domains in the case $n = 4$

Theorem (Morera's Theorem)

Let f be continuous in a domain Ω . If

$$\oint_\Gamma f = 0$$

along every simple closed contour Γ in Ω , then $f \in \mathcal{H}(\Omega)$.

Theorem (The Fundamental Theorem of Contour Integration (Second Form))

Let $f \in \mathcal{H}(\Omega^l)$ and let z_0 be a point in Ω^l . If Γ is a contour joining z_0 to z , then the function defined as

$$F(z) = \int_{z_0}^z f$$

is in $\mathcal{H}(\Omega^l)$. In this case, we have $F'(z) = f(z)$ for each $z \in \Omega^l$.

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Example

Find a closed form expression for the integral

$$\oint_{\Gamma} dz z^n$$

where $n \in \mathbb{Z}$ and Γ is the positively oriented circle centred at the origin and with radius r .

Theorem (Cauchy's Integral Formula)

Let $f \in \mathcal{H}(\Omega^l)$ and let Σ be a closed contour in Ω^l , if $z_0 \in \text{int } \Sigma$, then

$$f(z_0) = \frac{1}{2\pi i} \oint_{\Sigma} dz \frac{f(z)}{z - z_0}.$$

Lemma

Suppose that f is holomorphic on an open subset containing $\overline{B_R(z_0)}$ and satisfies $|f(z)| \leq M$ for all $z \in \overline{B_R(z_0)}$. Then for $0 < |z - z_0| < R/2$ the following are valid:

- i) $f'(z_0) = \frac{1}{2\pi i} \oint_{|\zeta - z_0|=R} d\zeta \frac{f(\zeta)}{(\zeta - z_0)^2};$
- ii) $\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| \leq 2M \frac{|z - z_0|}{R^2}.$

Theorem (Leibniz Rule)

Let γ be a path and let Ω be an open subset of $\mathbb{C}$. Let $\phi(z, \zeta)$ be a function defined for $z \in \Omega$ and $\zeta \in \gamma^\dagger$. Suppose that $\phi(z, \zeta) : \Omega \times \gamma^\dagger$ is continuous in $\gamma^\dagger$ and holomorphic in Ω with a continuous derivative $\partial_z \phi(z, \zeta)$ in $\gamma^\dagger$. Then the function

$$g(z) = \int_\gamma d\zeta \phi(z, \zeta)$$

is in $\mathcal{H}(\Omega)$ with derivative given by

$$\partial_z g(z) = \int_\gamma d\zeta \partial_z \phi(z, \zeta).$$

Theorem (Generalised Cauchy's Integral Formula)

Let f be holomorphic in a simply connected domain containing the simple closed contour Γ . Then f is infinitely differentiable at each $z \in \text{int } \Gamma$, with

$$\frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \oint_{\Gamma} dz \frac{f(z)}{(z - z_0)^{n+1}}.$$

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Remark

We can now complete the proof of the Cauchy-Riemann theorem, the following theorem is the remaining proof.

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Theorem

If f is holomorphic in a domain Ω , then all partial derivatives of u and v exist and are continuous in Ω .

Definition

Let Ω be an open subset of $\mathbb{C}$ and let $z_0 \in \Omega$. A function $f : \Omega \rightarrow \mathbb{C}$ is said to be **analytic at** z_0 if there exists a power series $\sum a_n(z - z_0)^n$ around z_0 with radius of convergence $r > 0$ such that $B_r(z_0) \subset \Omega$ and

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n \quad \text{for all } z \in B_r(z_0).$$

Further, we say that f is **analytic** if it is analytic at each point of Ω . Furthermore, the family of all analytic functions on Ω will be denoted by $\mathcal{A}(\Omega)$.

Theorem (Taylor's Theorem)

Let Ω be a domain, let $f \in \mathcal{H}(\Omega)$ and let z_0 be a point in Ω . Then $f \in \mathcal{A}(\Omega)$ with radius of convergence $R = \inf\{|z - z_0| : z \in \partial\Omega\}$, namely

$$f(z) = \sum_{n \in \mathbb{N}_0} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n, \quad \text{for } |z - z_0| < R.$$

This series representation is called the **Taylor expansion of f around z_0** .

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Corollary

If Ω is a domain in $\mathbb{C}$, then $\mathcal{H}(\Omega) = \mathcal{A}(\Omega)$.

Theorem (Cauchy's Inequality)

Let Γ be the positively oriented circle with centre z_0 and radius r and suppose f that is holomorphic inside and on Γ . If $|f(\Gamma)| \leq M$ for some positive number M , then

$$|f^{(n)}(z_0)| \leq \frac{Mn!}{r^n}, \quad n \in \mathbb{N}_0.$$

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Theorem (Liouville's Theorem)

If f is a bounded and entire function, then it must be a constant.

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Theorem (The Fundamental Theorem of Algebra)

If $P(z)$ is a nonconstant polynomial, then there exists z_0 such that $P(z_0) = 0$.